Varieties of Semiring Solvers via Multicategorical and Algebraic Constructions

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Introduction

Problem

Prove formally that $\forall a, b \in \mathbb{R}, (a+b)^2 = a^2 + 2ab + b^2$.

Formal proof: 10 axiom invocations.

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Automatisation:

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Require Import Ring.

Goal forall a b : R, (a + b)^2 = a^2 + 2*a*b + b^2.
Proof. intro a b. ring. Qed.
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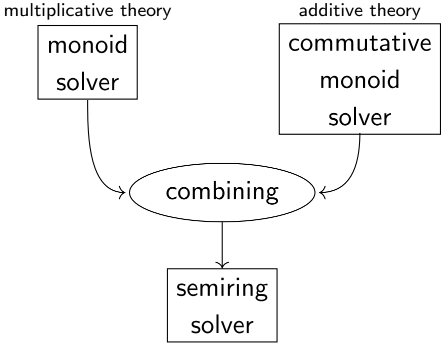
Objective


multiplicative theory
monoid solver
additive theory
commutative monoid solver
combining
semiring solver

Contributions

Outline

1. Preliminaries: universal algebra and the distributive combination of theories
2. Contribution: free algebra for the distributive combination of theories

Formal syntax

Finitary signature $\Sigma = (\mathrm{op}\,\Sigma,\ \mathrm{arity}_\Sigma)$
describes the operators of a theory

- *operator* set: $\mathrm{op}\,\Sigma \in \mathbf{Set}$
- *arity* assignment: $\mathrm{arity}_\Sigma : \mathrm{op}\,\Sigma \to \mathbb{N}$

Example: monoid signature $\Sigma_{\mathbf{Mon}}$

- $\mathrm{op}\,\Sigma_{\mathbf{Mon}} := \{(\cdot), 1\}$
- $\mathrm{arity}_{\Sigma_{\mathbf{Mon}}}(\cdot) := 2$, i.e, $(\cdot)$ is a binary operator
- $\mathrm{arity}_{\Sigma_{\mathbf{Mon}}}\, 1 := 0$, i.e, 1 is a constant

Written succintly: $\Sigma_{\mathbf{Mon}} := \{(\cdot) : 2,\ 1 : 0\}$

Interpretation

Algebra $A = (\underline{A},\ A[\![-]\!])$ over a signature Σ

- *carrier* set: $\underline{A} \in \mathbf{Set}$
- *operations* or *interpretations*: $A[\![f]\!] : \underline{A}^n \to \underline{A}$ for all $f : n \in \Sigma$

Example: monoids

- $(\mathbb{N}, (+), 0)$
- $(\mathbb{N}, (\cdot), 1)$
- $(\mathbb{N}, \max, 0)$
- $(\mathbb{C}, (+), 0)$
- $(\mathbb{C}, (\cdot), 1)$

Structure-preserving maps

Homomorphism $h : A \to B$

between Σ-algebras A, B:

- function $\underline{h} : \underline{A} \to \underline{B}$
- such that:

$$B\,[\![f]\!]\big(h(a_1), \ldots, h(a_n)\big) = h\big(A\,[\![f]\!]\,(a_1, \ldots, a_n)\big)$$

for all $f : n \in \Sigma$ and $a_1, \ldots, a_n \in \underline{A}$

Example: complex conjugation

- $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}, \quad \overline{0} = 0$
- $\overline{z_1 \cdot z_2} = \overline{z_1} \cdot \overline{z_2}, \quad \overline{1} = 1$

Abstract syntax

$\mathrm{Term}_\Sigma X$: terms over signature Σ and variable set X

$$\frac{x \in X}{x \in \mathrm{Term}_\Sigma X} \qquad \frac{t_i \in \mathrm{Term}_\Sigma X \quad \mathrm{op} : n \in \Sigma}{\mathrm{op}(t_1, \ldots, t_n) \in \mathrm{Term}_\Sigma X}$$

Term folding $A[\![t]\!]$

$$\left.\begin{array}{l} \blacktriangleright\ \Sigma\text{-algebra } A \\ \blacktriangleright\ \textit{environment } e : X \to \underline{A} \end{array}\right\} \Sigma - \textit{algebra over } X$$

Inductively:

$$A[\![x]\!]\, e := e\, x \qquad A[\![\mathrm{op}(t_1, \ldots, t_n)]\!]\, e := A[\![\mathrm{op}]\!] \left(A[\![t_1]\!]\, e, \ldots, A[\![t_n]\!]\, e\right)$$

Example: monoid

- term $t := x \cdot (y \cdot 1)$
- interpretation with $e := \{x \mapsto 2,\ y \mapsto 3\}$ in $A := (\mathbb{N}, (+), 0)$

$$A[\![t]\!] = 2 + (3 + 0) = 5$$

Validity

Equation $X \vdash_\Sigma l = r$ over signature Σ

- set X
- terms $l, r \in \mathrm{Term}_\Sigma X$

Satisfaction $\langle A, e \rangle \vDash (X \vdash l = r)$
by Σ-algebra over X $\langle A, e \rangle$:

$$A \llbracket l \rrbracket e = A \llbracket r \rrbracket e$$

Validity $A \vDash (X \vdash l = r)$
in algebra A:

$$\forall e : X \to \underline{A}, \quad \langle A, e \rangle \vDash (X \vdash l = r)$$

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Validity $A \vDash (X \vdash l = r)$
in algebra A:

$$\forall e : X \to \underline{A}, \quad \langle A, e \rangle \vDash (X \vdash l = r)$$

Example:

- $\langle A \coloneqq (\mathbb{N}, (+), 0),\ e \coloneqq \{x \mapsto 2,\ y \mapsto 3\}\rangle$ satisfies

  $$\{x, y\} \vdash x \cdot (y \cdot 1) = 1 \cdot (x \cdot y)$$

  since

  $$2 + (3 + 0) = 5 = 0 + (2 + 3)$$

Varieties of algebras

Presentation $\mathcal{T} = (\Sigma_{\mathcal{T}}, \mathrm{Ax}_{\mathcal{T}})$

- signature $\Sigma_{\mathcal{T}}$
- set $\mathrm{Ax}_{\mathcal{T}}$ of $\Sigma_{\mathcal{T}}$-equations, the *axioms*

$\mathcal{T}$*-model*: $\Sigma_{\mathcal{T}}$-algebra validating every axiom in $\mathcal{T}$: $A \vDash (X \vdash l = r) \in \mathrm{Ax}_{\mathcal{T}}$

Example: monoids

Monoid presentation := $(\Sigma_{\mathbf{Mon}}, \mathrm{Ax}_{\mathbf{Monoid}})$ with the axioms

$$\{x, y, z\} \vdash x \cdot (y \cdot z) = (x \cdot y) \cdot z \qquad \text{associativity}$$

$$\{x\} \vdash 1 \cdot x = x \qquad \text{left neutrality}$$

$$\{x\} \vdash x \cdot 1 = x \qquad \text{right neutrality}$$

monoids are **Monoid**-models

Free algebras

Free $\mathcal{T}$-model over X $\langle F_{\mathcal{T}}X,\ \eta\rangle$

- $\mathcal{T}$-model $F_{\mathcal{T}}X$
- environment η called the *unit*
- *universal property*:
  every $\mathcal{T}$-model over X $\langle A,\ e\rangle$ uniquely defines
  - a homomorphism $h : F_{\mathcal{T}}X \to A$
  - that extends e along η: $e = h \circ \eta$

Example: free monoid over X

- $\underline{F_{\mathbf{Monoid}}X} := \mathrm{List}\,X \qquad u \cdot v := u \mathbin{+\!\!+} v \qquad 1 := [\,] \qquad \eta\, x := [x]$
- for every **Monoid**-model over X $\langle A, e\rangle$, a unique monoid homomorphism h:

$$\underline{h}[x_1, \ldots, x_n] := 1 \cdot (e\ x_1) \cdot \ldots \cdot (e\ x_n)$$

$h(\eta\ x) = h[x] = 1 \cdot (e\ x) = e\ x$ since $F_{\mathbf{Monoid}}X$ is a **Monoid**-model

Free algebras and normalisation

Theorem (extensional normalisation)

Let $\langle F_{\mathcal{T}}X, \mathrm{env}\rangle$ be a free $\mathcal{T}$-model, and let $X \vdash l = r$ be an equation in context. The following are equivalent:

1. *The unit satisfies the equation: $\langle F_{\mathcal{T}}X, \mathrm{env}\rangle \vDash (X \vdash l = r)$*
2. *The free algebra validates the equation: $F_{\mathcal{T}}X \vDash (X \vdash l = r)$*
3. *Every $\mathcal{T}$-model A validates the equation: $A \vDash (X \vdash l = r)$*

Proof

3. $\Rightarrow$ 2. and 2. $\Rightarrow$ 1. hold by definition.

1. $\Rightarrow$ 3. is deduced from the universal property of the free algebra.

$\square$

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Example: monoids

The equation

$$\{x, y\} \vdash x \cdot (y \cdot 1) = 1 \cdot (x \cdot y)$$

holds in every monoid since

$$[x] +\!\!+ ([y] +\!\!+ []) = [x,\ y] = [] +\!\!+ ([x] +\!\!+ [y])$$

The distributive combination of theories

Additive part

Presentation $\mathcal{A} = \langle \Sigma_{\mathcal{A}}, \mathrm{Ax}_{\mathcal{A}} \rangle$

Multiplicative part

Presentation $\mathcal{M} = \langle \Sigma_{\mathcal{M}}, \mathrm{Ax}_{\mathcal{M}} \rangle$

Distributive combination $\mathcal{M} \triangleright \mathcal{A}$

$\Sigma_{\mathcal{M} \triangleright \mathcal{A}} = \Sigma_{\mathcal{A}} \sqcup \Sigma_{\mathcal{M}}$

$\mathrm{Ax}_{\mathcal{M} \triangleright \mathcal{A}} = \mathrm{Ax}_{\mathcal{A}} \sqcup \mathrm{Ax}_{\mathcal{M}}$

$\sqcup$ distributivity axioms of every $\mathcal{M}$-operation over every $\mathcal{A}$-operation

Example

Semiring $=$ monoid $\triangleright$ commutative monoid

Distributivity axioms:

$$x, y, z \vdash x \cdot (y + z) = (x \cdot y) + (x \cdot z) \qquad x, y, z \vdash (x + y) \cdot z = (x \cdot z) + (y \cdot z)$$

$$x \vdash x \cdot 0 = 0 \qquad x \vdash 0 \cdot x = 0$$

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Commutative theories

Commutative theory

- $\mathcal{T}$ presentation
- such that every operator commutes with every other operator:

$$f\begin{pmatrix} g(x_{11},\ldots,x_{1m}) \\ \vdots \\ g(x_{n1},\ldots,x_{nm}) \end{pmatrix} = g\left(f\begin{pmatrix} x_{11} \\ \vdots \\ x_{n1} \end{pmatrix} \quad \cdots \quad f\begin{pmatrix} x_{1m} \\ \vdots \\ x_{nm} \end{pmatrix} \right)$$

holds in every $\mathcal{T}$-model for every $f : n \in \Sigma_{\mathcal{T}}$, $g : m \in \Sigma_{\mathcal{T}}$

Example: monoids

The equations are:

$$(x \cdot y) \cdot (z \cdot w) = (x \cdot z) \cdot (y \cdot w) \qquad 0 \cdot 0 = 0 \qquad 0 = 0$$

The theory of commutative monoids is commutative.

Failure: no-go results

[Zwart & Marsden, 2018]: 3 classes of no-go results

1. $\mathcal{M}$ has an idempotent operator
   (i.e. there exists $* : 2 \in \Sigma_{\mathcal{M}}$ such that $x \vdash x * x = x$ holds)
2. $\mathcal{A}$ has two or more distinct constants
3. $\{a, b, c, d\} \vdash (a * b) * (c * d) = (a * c) * (b * d)$ does not hold for every $* : 2 \in \Sigma_{\mathcal{A}}$

2. and 3. are false under the commutativity hypothesis

Problem

What about 1.?

Affine theories

Affine theory

- $\mathcal{T}$ presentation
- such that in every axioms, each variable appears at most once

Examples:

- $\{x, y\} \vdash x + y = y + x$ is affine
- $\{x\} \vdash x \cdot 1 = x$ is affine
- $\{x\} \vdash x \vee x = x$ is *not* affine

Semigroups, monoids, groups: all affine theories

Conclusion

- free algebras:

  normal forms modulo axioms

- distributive combination of theories:

  disjoint union + distributivity axioms

- free algebra for the distributive combination of theories:
  composition of the component free algebras
  - commutative additive theory
  - affine multiplicative theory

- modularity:

  already 28 non-trivial solvers

Section 3

Appendices

Extensions

Free extensions